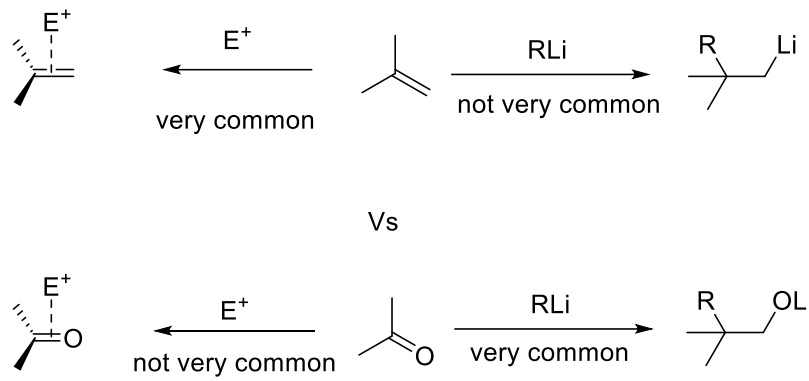
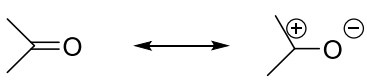
Reactions of carbonyl compounds

Difference in reactivity patterns



Carbonyl compounds are electrophilic.

1. Charge:

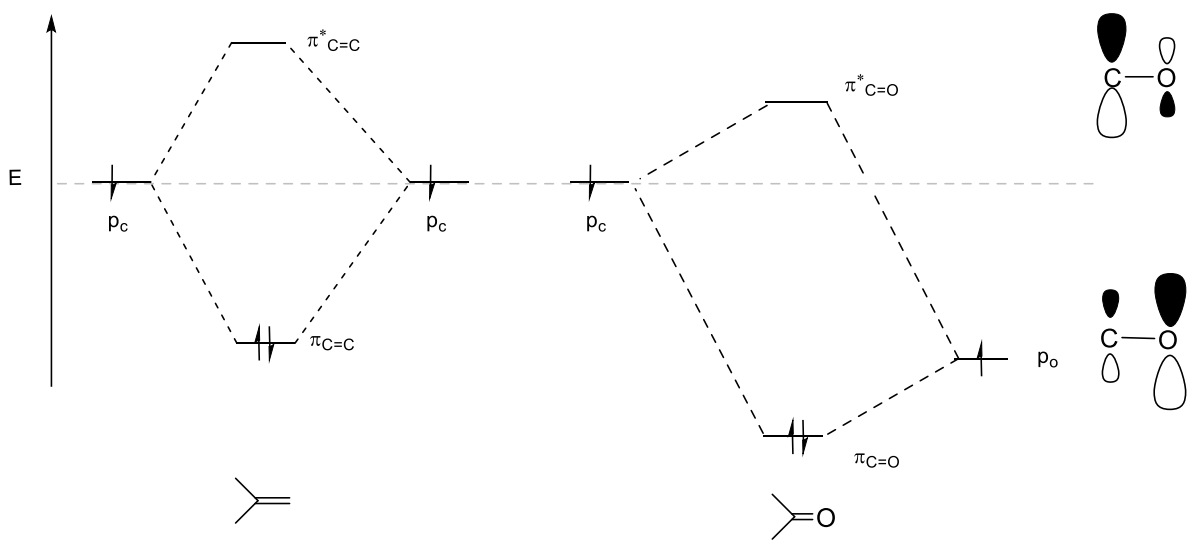


μ = 2.3 D

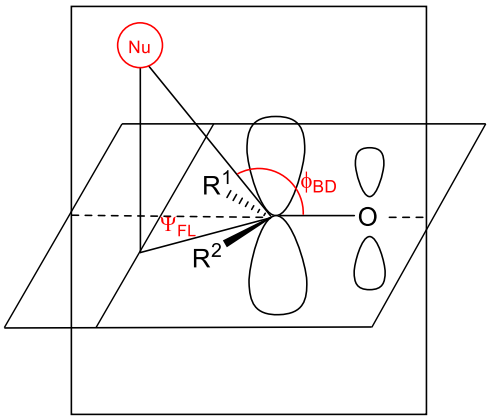
- Dipole moment μ = 2.8 D
- If C^+ and O^- are considered as point charges, then μ = q × d = 5.9 D
- % ionic character = 2.8/5.9 × 100 % = 47%
- significant contribution of the charge separated resonance structure
- The high reactivity of carbonyl compounds thus can be associated with both MO and charge.

2. MO

$\pi^*_{C=O}$ is lower in energy and C - center has bigger coefficienct



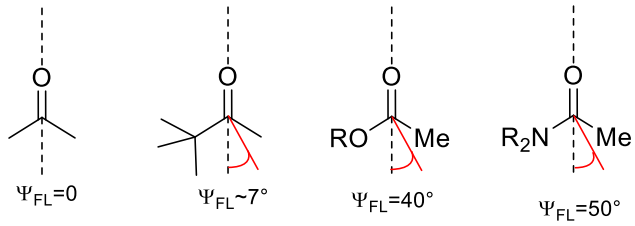
Stereoelectronic effect during the addition of a nucleophile to the carbonyl compound $\pi^*_{C=O}$



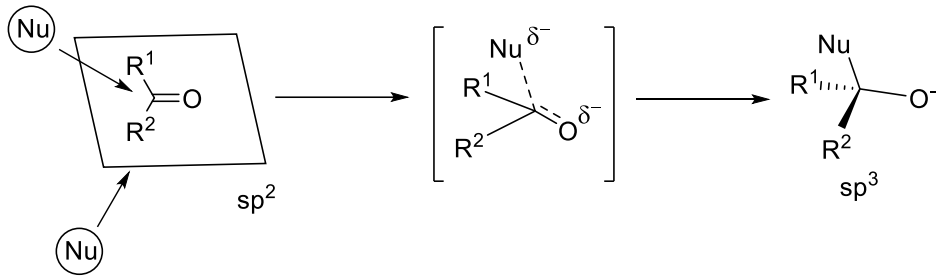
ϕ_{BD} = Burgi – Dunitz trajectory

~ 107 ± 5°

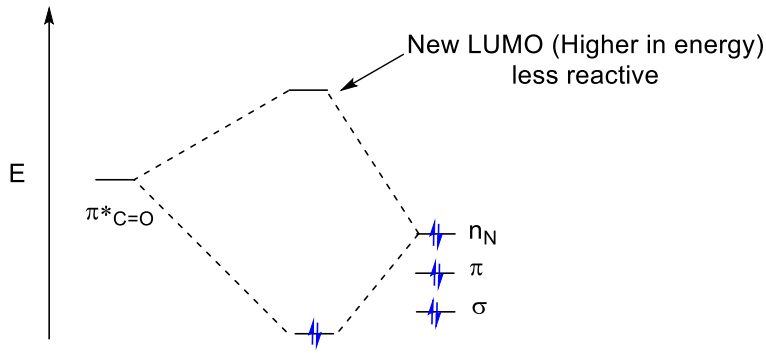
Ψ_{FL} = Flippin – Lodge angle



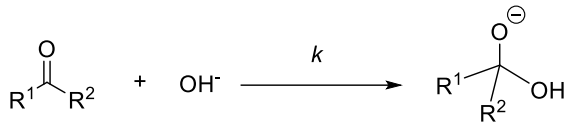
General reactivity of carbonyl compounds



- attack from both side possible.
- Sp^2 to sp^3 sterics is increasing in the TS
- R^1 and R^2 groups reduces the reactivity of $\pi^*_{C=O}$
  1. Steric ground (sp^2 to sp^3)
  2. Electronics ground (MO)



Example:



k_{rel} =	1	10^{-2}	10^{-4}	10^{-7}	10^{-11}	$< 10^{-11}$

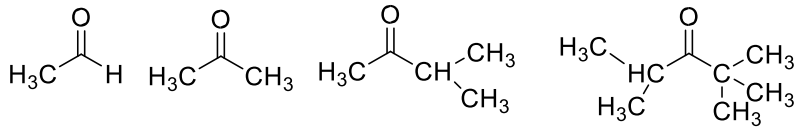
- sterics
- $\sigma_{C-H} \rightarrow \pi^*_{C=O}$

- sterics
- $\sigma_{C-H} \rightarrow \pi^*_{C=O}$ (more)

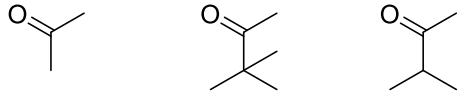
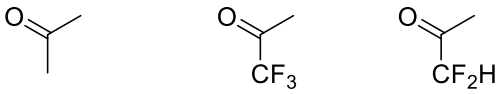
- sterics
- $\sigma_{C-H} \rightarrow \pi^*_{C=O}$
- $n_o \rightarrow \pi^*_{C=O}$ more contributing

Questions:

1. Write structural formulas for all ketones with the molecular formula $C_6H_{12}O$ and give each its IUPAC name. Which of these ketones are chiral?
2. Racemic 2-phenyl-2-butanol can be synthesized by three different combinations of a Grignard reagent and a ketone. Show each combination.
3. Based on purely electron ability, which C=O bond would undergo a faster reaction with nucleophiles – aldehydes or ketones?
4. Rate the addition of Nucleophile to each of the following aldehyde/ ketone?



5. State the order of reactivity of water with different carbonyls such as Acetone, Acetaldehyde, Formaldehyde and Ethyl Acetate with proper reasoning and mention product.
6. Why do carbonyl compounds act as electrophiles? Explain in terms of MO and charge.
7. Rank the following carbonyl compounds from most reactive to least reactive towards nucleophilic addition. Explain your reasoning.

a. 

b.